UNIT 2 · STANDARD 6.DS.5

# Represent and Describe Data in a Box Plot

Lesson 2-4

**Name:** _____ **Date:** _____ **Class:** _____

## My Notes

**Level 3 Enrichment**

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can make and read a box plot to summarize a data set.**
**Language Objective: I can explain my box plot using the words box plot, median, quartile, and interquartile range.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

**Box Plot** · **Median** · **Quartile** · **Interquartile Range** · **Data distribution** · **Variability** · **Five-number summary** · **Whisker** · **Middle half**

*Level 3: try the blanks from memory first, then check the bank.*

**1** A graph that shows data with a box and whiskers based on the five-number summary is a _____ .

**2** The middle value of the data, shown as a line inside the box, is the _____ .

3. A value that divides the data into four equal parts is a _____ .

4. The distance between the first and third quartiles, showing the middle 50% of data, is the _____ .

5. The way data values are spread out is the _____ .

6. How spread out the data values are is the _____ .

7. The minimum, Q1, median, Q3, and maximum together are the _____ .

8. The line that reaches from the box to an extreme value is a _____ .

9. The box of a box plot covers the _____ of the data.

## Write About the Math

**Which team is more consistently scoring high? Use the box plot values to support your answer?**
*¿Qué equipo anota alto de forma más consistente? Usa los valores del diagrama de caja para apoyar tu respuesta?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Box Plot** *Diagrama de caja*
- ☐ **Median** *Mediana*
- ☐ **Quartile** *Cuartil*
- ☐ **Interquartile Range** *Rango intercuartílico*
- ☐ **Data distribution** *Distribución de datos*

**Model:** *A box plot is built from the minimum, Q1, median, Q3, and maximum.* — Students name the five-number summary (min, Q1, median, Q3, max) and find the median (14) for the 11 ordered scores.

## Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Team A's IQR is ___ and Team B's IQR is ___. Team B's middle 50% scores between ___ and ___. The fan is wrong because ___. The box plot shows that Team B is more consistent because its Q1 and median are higher and its IQR (10) is smaller, while ___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

## Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.